



Preface

WinUI 3 is the newest desktop UI framework for Windows application development. It is part of Microsoft's Windows App SDK, providing developers with the tools to build beautiful apps with the Fluent Design System. This book will quickly get you up to speed with WinUI to build new Windows applications and to build apps across platforms with technologies such as Blazor and Uno Platform.

The book begins by exploring the history of Windows UI development frameworks to gain an understanding of how earlier frameworks influenced WinUI as it exists today. It covers the basics of XAML-based UI development and explores the controls available in WinUI before moving on to an examination of patterns and best practices for WinUI developers. To help reinforce these concepts, the early chapters in the book build practical skills by creating an application to organize a collection of books, music, and movies. Each chapter enhances the application, with new controls and concepts discussed.

Later chapters in the book explore how developers can leverage their WinUI knowledge to leverage open source toolkits, integrate web content in Windows apps, and migrate WinUI apps to Android, iOS, and the web. The book finishes by teaching some essential Visual Studio debugging techniques and discussing app deployment options to get your apps in the hands of consumers and enterprise users. At the end of every chapter, I've included a series of questions for you to attempt on your own, enabling you to gauge your level of understanding. Learn how WinUI can help you build and deploy modern, robust applications!

Who this book is for

This book is for anyone who wants to develop Windows applications with a modern **user experience (UX)**. If you are familiar with Windows Forms, UWP, or WPF, and are looking to update your knowledge of Windows development or modernize existing apps, this book is for you. If you are just learning .NET development, you can take advantage of this book to learn the basics of XAML development in parallel with your C# and .NET journey.

What this book covers

Chapter 1, *Introduction to WinUI*, examines the history of UI frameworks in Windows and the origins of WinUI, and you will create your first WinUI 3 project in Visual Studio.

Chapter 2, *Configuring the Development Environment and Creating the Project*, explains how to install and configure Visual Studio for WinUI development, the basics of XAML and C#, and kicks off the hands-on with a project that will be enhanced throughout the book.

Chapter 3, *MVVM for Maintainability and Testability*, introduces the basics of Model-View-ViewModel (MVVM) pattern, one of the most important design patterns when building XAML-based applications.

Chapter 4, *Advanced MVVM Concepts*, builds on the basics you learned about the MVVM pattern in WinUI apps to handle more advanced techniques. You'll learn how to keep components loosely coupled and testable when adding new dependencies to your projects.

Chapter 5, *Exploring WinUI Controls*, explores some of the many controls and APIs that WinUI offers for developers building Windows applications. This chapter explores the brand-new controls and updated controls that were previously available in WinUI 2 and UWP.

Chapter 6, Leveraging Data and Services, looks at managing data, a core part of software development. This chapter covers some key concepts of data management, including state management and the service locator pattern.

Chapter 7, Fluent Design System for Windows Applications, explains the tenets of Microsoft's Fluent Design System and how to implement them in your WinUI applications.

Chapter 8, Adding Windows Notifications to WinUI Applications, covers how to leverage the Windows App SDK to support push notifications and app notifications in your WinUI applications.

Chapter 9, Enhancing Applications with the Windows Community Toolkits, introduces the Windows Community Toolkit and the .NET Community Toolkit – collections of open source libraries for Windows developers. You will learn how to leverage the controls and helpers from the toolkits in your WinUI projects.

Chapter 10, Accelerating App Development with Template Studio, shows how to leverage Template Studio to create a new WinUI project, which can be a daunting task, built on the best Windows development patterns and practices.

Chapter 11, Debugging WinUI Applications with Visual Studio, shows how to leverage the XAML debugging tools in Visual Studio to track down pesky bugs in your WinUI project – good debugging skills are essential for developers.

Chapter 12, Hosting a Blazor Application in WinUI, looks at the WebView2 control in WinUI and using it to host a Blazor application deployed to the cloud from inside your Windows application.

Chapter 13, Take Your App Cross-Platform with Uno Platform, explains how to migrate a WinUI project to Uno Platform, which allows devel-

opers to write XAML and C# code in a single code base and run it on any platform.

Chapter 14, *Packaging and Deploying WinUI Applications*, explores some of the multiple options WinUI developers have for packaging and deploying WinUI applications, looking at deploying through the Microsoft Store, WinGet, and side-loading apps.

To get the most out of this book

If you are familiar with Windows Forms, .NET MAUI, UWP, or WPF and are looking to enhance your knowledge of Windows development or modernize existing apps, you will find this book useful. Hands-on experience with C# and .NET is expected but no prior knowledge of WinUI is required.

Software/hardware covered in the book	Operating system requirements
WinUI 3	Windows 10 version 1809 or later or Windows 11
C#	Windows, macOS, or Linux
.NET 7	Windows, macOS, or Linux
Visual Studio 2022	Windows 10 or 11
Blazor	Windows, macOS, or Linux
Uno Platform	Windows, macOS, or Linux

The book covers how to get started with WinUI development, but you should have Visual Studio and .NET installed. Follow the instructions on Microsoft Learn:

<https://learn.microsoft.com/visualstudio/install/install-visual-studio>.

If you are using the digital version of this book, we advise you to type the code yourself or access the code from the book's GitHub repository (a link is available in the next section). Doing so will help you avoid any potential errors related to the copying and pasting of code.

After you read this book, you can continue your Windows development journey by diving deeper into the documentation and samples on Microsoft Learn: <https://learn.microsoft.com/windows/apps/>.

Download the example code files

You can download the example code files for this book from GitHub at <https://github.com/PacktPublishing/Learn-WinUI-3-Second-Edition>.

If there's an update to the code, it will be updated in the GitHub repository.

We also have other code bundles from our rich catalog of books and videos available at <https://github.com/PacktPublishing/>. Check them out!

Conventions used

There are a number of text conventions used throughout this book.

Code in text: Indicates code words in text, database table names, folder names, filenames, file extensions, pathnames, dummy URLs, user input, and Twitter handles. Here is an example: “In

INavigationService, you can update the **namespace** to **UnoMediaCollection.Interfaces** and remove the **using System;** statement.”

A block of code is set as follows:

```
using UnoMediaCollection.Enums;  
using UnoMediaCollection.Interfaces;  
using UnoMediaCollection.Model;
```

When we wish to draw your attention to a particular part of a code block, the relevant lines or items are set in bold:

```
using UnoMediaCollection.Enums;  
namespace UnoMediaCollection.Model
```

Any command-line input or output is written as follows:

```
$ mkdir css  
$ cd css
```

Bold: Indicates a new term, an important word, or words that you see onscreen. For instance, words in menus or dialog boxes appear in **bold**. Here is an example: “Right-click the **Enums** folder and select **Add | Existing Item**.”

TIPS OR IMPORTANT NOTES

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